Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 30: Approved Solutions

Recursive Sequences and Strong Induction

Covers Chartrand & Zhang, Sections 4.3 and 4.4

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Ordinary or strong?

Ask: **how far back does the inductive step need to reach?**

- Only to $k \Rightarrow$ ordinary induction.
- To $k-1$, $k-2$, or somewhere unpredictable $\Rightarrow$ strong induction.

If the step reaches back j places, you need j consecutive base cases.

Three mistakes to expect

1. **Using a recurrence where it does not apply.** Check the "for $n \geq \ldots$" condition.
2. **Too few base cases.** If the step uses $k-3$, you need four starting values.
3. **Conjecturing from too few terms.** Compute at least four or five before guessing.

Part II: Practice Problems

Section A — Recursive sequences

1. Let $a_1 = 1$ and $a_n = a_{n-1} + 2n - 1$ for $n \geq 2$.

(a) Compute a_2 through a_5 and conjecture a formula.

(b) Prove your formula.

Approved solution

(a) $a_2 = 1 + 3 = 4$, $a_3 = 4 + 5 = 9$, $a_4 = 9 + 7 = 16$, $a_5 = 16 + 9 = 25$. Conjecture: $a_n = n^2$.
(b) Ordinary induction suffices, since each term uses only the previous one. Base: $a_1 = 1 = 1^2$.

Assume $a_k = k^2$. Then

$$a_{k+1} = a_k + 2(k+1) - 1 = k^2 + 2k + 1 = (k+1)^2.$$

By induction $a_n = n^2$ for all n. ■

This is Result 4.4 in disguise — the sum of the first n odd numbers. The recurrence adds the next odd number at each step.

Section B — Strong induction

2. Let $a_1 = 1$, $a_2 = 2$, and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$. Prove that $a_n < 2^n$ for every positive integer n.

Approved solution

Proof. We use strong induction, since each term depends on the two before it.
Base steps. $a_1 = 1 < 2$ and $a_2 = 2 < 4$. ✓
Inductive step. Let $k \geq 2$ and assume $a_i < 2^i$ for every i with $1 \leq i \leq k$. Since $k+1 \geq 3$, the recurrence applies:

$$a_{k+1} = a_k + a_{k-1} < 2^k + 2^{k-1} < 2^k + 2^k = 2^{k+1}.$$

By strong induction, $a_n < 2^n$ for all n. ■

Two base cases, because the step reaches back two places. With only a_1 checked, the step for $k = 1$ would need a_0, which does not exist. That is Chartrand's Result 4.30 warning.

3. Prove that every amount of postage of 8 cents or more can be made using only 3-cent and 5-cent stamps.

Approved solution

Proof. We use strong induction.
Base steps. $8 = 3 + 5$, $9 = 3 + 3 + 3$, $10 = 5 + 5$. ✓
Inductive step. Let $k \geq 10$ and assume every amount from 8 to k can be made. We show $k+1$ can. Since $k + 1 \geq 11$, we have $(k+1) - 3 = k - 2 \geq 8$, so $k - 2$ can be made by the inductive hypothesis. Add one 3-cent stamp. ■

Three base cases, because the step reaches back three places (to $k - 2$). And the threshold is sharp: 7 cents cannot be made from 3s and 5s.

4. Prove that every integer $n \geq 2$ is either prime or a product of primes.

Approved solution

Proof. We use strong induction, starting at 2.
Base step. 2 is prime. ✓
Inductive step. Let $k \geq 2$ and assume every integer i with $2 \leq i \leq k$ is prime or a product of primes. Consider $k+1$.
Case 1: $k+1$ is prime. Then we are done.
Case 2: $k+1$ is not prime. Then $k + 1 = ab$ with $2 \leq a, b \leq k$. By the inductive hypothesis, each of a and b is prime or a product of primes, so ab is a product of primes.
By strong induction the result holds for all $n \geq 2$. ■

This is the clearest reason strong induction exists. The factors a and b can be *anywhere* below $k+1$ — you cannot predict which earlier case you will need, so you assume them all. Ordinary induction, which gives only $P(k)$, would be stuck.

Section C — Writing

5. In five to eight complete sentences, explain why a recursive function can be proved correct by strong induction. Describe what plays the role of the base step and the inductive hypothesis, and say what bug corresponds to a missing base case.

Approved solution

Model answer. A recursive function computes its answer for an input by calling itself on smaller inputs and combining the results. Its correctness for all inputs can be established by strong induction on the input size. The base step corresponds to the function's base cases: for the smallest inputs, it returns an answer directly, and one checks those answers are right. The inductive hypothesis corresponds to assuming that every recursive call on a smaller input returns the correct result; the inductive step then checks that combining those correct sub-results gives the correct answer for the current input. Strong induction is the right form because a recursive call may reduce the input by any amount, not just by one. A missing base case in the proof matches a missing base case in the code, where the recursion never stops — the same failure, since in both cases the chain of reasoning has nothing real to rest on.

Full credit requires matching base cases to base step and recursive calls to the inductive hypothesis, plus the missing-base-case point.

Instructor notes

- **Problem 4 is the argument for strong induction.** The factors can be anywhere below $k+1$; ordinary induction cannot reach them. Worth board time.
- **Problem 5** is the CS connection — recursive correctness is strong induction. Students who get this understand why the course covers induction at all.
- **Problems 2 and 3** teach the rule: reach back j places, need j base cases.
- **Problem 1** is Result 4.4 hidden in a recurrence.
- Reserved for Homework 11: Sections 4.3 and 4.4 even-numbered exercises.